

Linear Algebra—Winter/Spring 2026

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Practice Set 1

1. Let

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 2 & -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \\ 1 & 4 & -2 \end{pmatrix}.$$

- Express each column of AB as a linear combination of the columns of A .
- Express each row of AB as a linear combination of the rows of B .

2. Solve the following system of linear equations:

$$\begin{aligned} x_1 - x_2 - x_3 &= 2, \\ 3x_1 - 3x_2 + 2x_3 &= 16, \\ 2x_1 - x_2 + x_3 &= 9. \end{aligned}$$

3. Let

$$A = \begin{pmatrix} 6 & -4 & 0 \\ 4 & -2 & 0 \\ -1 & 0 & 3 \end{pmatrix}.$$

Find all vectors $X \in \mathbb{R}^3$ that satisfy

$$AX = 3X.$$

4. Describe explicitly all 3×3 row-reduced echelon matrices.

5. Recall that the existence of an additive inverse is usually included as one of the axioms in the definition of a vector space. Show that this condition can be replaced by the requirement that

$$0v = 0 \quad \text{for all } v \in V,$$

where 0 on the left-hand side is the scalar 0 and the 0 on the right-hand side is the additive identity of V .

Hint. Use this requirement to show the existence of an additive inverse for each vector $v \in V$.

- Let V be a vector space over a field F . Show that the singleton set $\{\alpha\} \subseteq V$ is linearly independent if and only if $\alpha \neq 0$.
- Suppose V is a finite-dimensional vector space and W is a subspace of V . Show that there is a subspace U of V such that $V = U + W$ and $U \cap W = \{0\}$.

8. Let V be the set of all 2×2 matrices with complex entries whose diagonal entries add up to zero. Determine whether V is a vector space over the field of complex numbers. Also determine whether V is a vector space over the field of real numbers.
9. The vectors $\alpha_1 = (1, 2)$ and $\alpha_2 = (3, 4)$ form a basis of $\mathbb{R}^2$. Find a linear transformation T from $\mathbb{R}^2$ into $\mathbb{R}^3$ such that

$$T(\alpha_1) = (2, 5, 2),$$

$$T(\alpha_2) = (4, 7, 0).$$

Also, find $T(1, 0)$ and $T(0, 1)$.

10. Verify the statement of the rank–nullity theorem for the following linear transformation:

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad T(x_1, x_2, x_3) = (x_1, x_2).$$